Elevator – Maintenance Practices

1. Removal/Installation

CAUTION **IF REMOVING OR INSTALLING CONTROL CABLES, DISCONNECT MAIN BATTERY (REFER TO 24-30-31, MAIN BATTERY SYSTEM – MAINTENANCE PRACTICES, PARA 2.), AVIONICS START BATTERY (IF INSTALLED, REFER TO 24-30-32, AVIONICS START BATTERY (MOD 6/2315) – GENERAL DATA, PARA 4.) AND EMERGENCY BATTERY (IF INSTALLED, REFER TO 33-50-01, EMERGENCY LIGHTING BATTERY – GENERAL DATA, PARA 1.) AND DISCONNECT GROUND POWER.**

A. Remove Left Elevator

1) Disconnect elevator trim tab system cables in rear fuselage.

2) Remove elevator trim tab system rear pulley assembly (refer to 27-33-00, Elevator Trim Tab System – General Data).

3) Remove three bolts from torque tube flanges. Disconnect bonding jumper (refer to Figure 1, Detail A).

4) Support elevator, remove bolt at STA 113.1 hinge arm (refer to Figure 1, Detail C). Disconnect bonding jumper.

WARNING **DO NOT REMOVE THE ELEVATOR BY REMOVING THE TWO SCREWS AT THE STA 113.1 HINGE ARM. REPEATED REMOVAL AND RE-INSTALLATION WILL DAMAGE THE ANCHOR NUT PLATES. THIS COULD RESULT IN CONTROL SURFACE JAMMING.**

5) Support elevator, remove two bolts at STA 56.3 hinge arm (refer to Figure 1, Detail B). Disconnect sensor wiring at WL 58.875.

6) Lower elevator to ground.

B. Install Left Elevator

CAUTION **ALL SELF-LOCKING NUTS PREVIOUSLY REMOVED ARE TO BE DISCARDED AND REPLACED WITH NEW. DO NOT RE-USE ANY SELF-LOCKING NUTS.**

Note If the elevator to be installed is a replacement or repaired elevator, or if it has been painted or repainted, rebalancing of elevators must be carried out in accordance with Structural Repair Manual PSM 1–6–3.

1) Position elevator, install and secure all hinge bolts (refer to Figure 1). Connect and secure bonding jumpers and sensor wiring at WL 58.875. Refer to PSM 1–64–2, Chapter 20 for bonding jumper requirements.

2) Install and secure three bolts to torque tube flanges (refer to Figure 1, Detail A). Connect and secure bonding jumper.

3) Install elevator trim tab system rear pulley (refer to 27-33-00, Elevator Trim Tab System – General Data).

4) Connect elevator trim tab system cables.

5) Rig elevator control system (refer to 27-30-00, Elevator Control System – General Data).

6) Do an elevator position operational test as per 27-30-21, Elevator Angle Position Indicator – General Data, Para 3.B.

7) If adjustment is required, adjust the elevator angle position indicator as per 27-30-21, Elevator Angle Position Indicator – General Data, Para 3.A.

C. Remove Right Elevator

1) Disconnect flap/elevator interconnect trim system cables in rear fuselage, or at quick disconnects as applicable.

2) Disconnect connecting rod from elevator torque tube.

3) Remove flap/elevator interconnect system rear pulley assembly (refer to 27-36-00, Flap/Elevator Interconnect Trim System – General Data).

4) Remove three bolts from torque tube flanges. Disconnect bonding jumper (refer to Figure 1, Detail A).

5) Support elevator, remove bolt at STA 113.1 hinge arm (refer to Figure 1, Detail C) and disconnect bonding jumper.

WARNING **DO NOT REMOVE THE ELEVATOR BY REMOVING THE TWO SCREWS AT THE STA 113.1 HINGE ARM. REPEATED REMOVAL AND RE-INSTALLATION WILL DAMAGE THE ANCHOR NUT PLATES. THIS COULD RESULT IN CONTROL SURFACE JAMMING.**

6) Support elevator, remove two bolts at STA 56.3 hinge arm (refer to Figure 1, Detail B). Disconnect sensor wiring at WL 58.875.

7) Lower elevator to ground.

D. Install Right Elevator

CAUTION **ALL SELF-LOCKING NUTS PREVIOUSLY REMOVED ARE TO BE DISCARDED AND REPLACED WITH NEW. DO NOT RE-USE ANY SELF-LOCKING NUTS.**

Note If the elevator to be installed is a replacement or repaired elevator, or if it has been painted or repainted, rebalancing of elevators must be carried out in accordance with Structural Repair Manual PSM 1–6–3.

1) Position elevator, install and secure all hinge bolts (refer to Figure 1). Connect and secure bonding jumpers and sensor wiring at WL 58.875. Refer to PSM 1–64–2, Chapter 20 for bonding jumper requirements.

2) Install and secure three bolts to torque tube flanges (refer to Figure 1, Detail A). Connect and secure bonding jumper.

3) Install flap/elevator interconnect system rear pulley assembly (refer to 27-36-00, Flap/Elevator Interconnect Trim System – General Data).

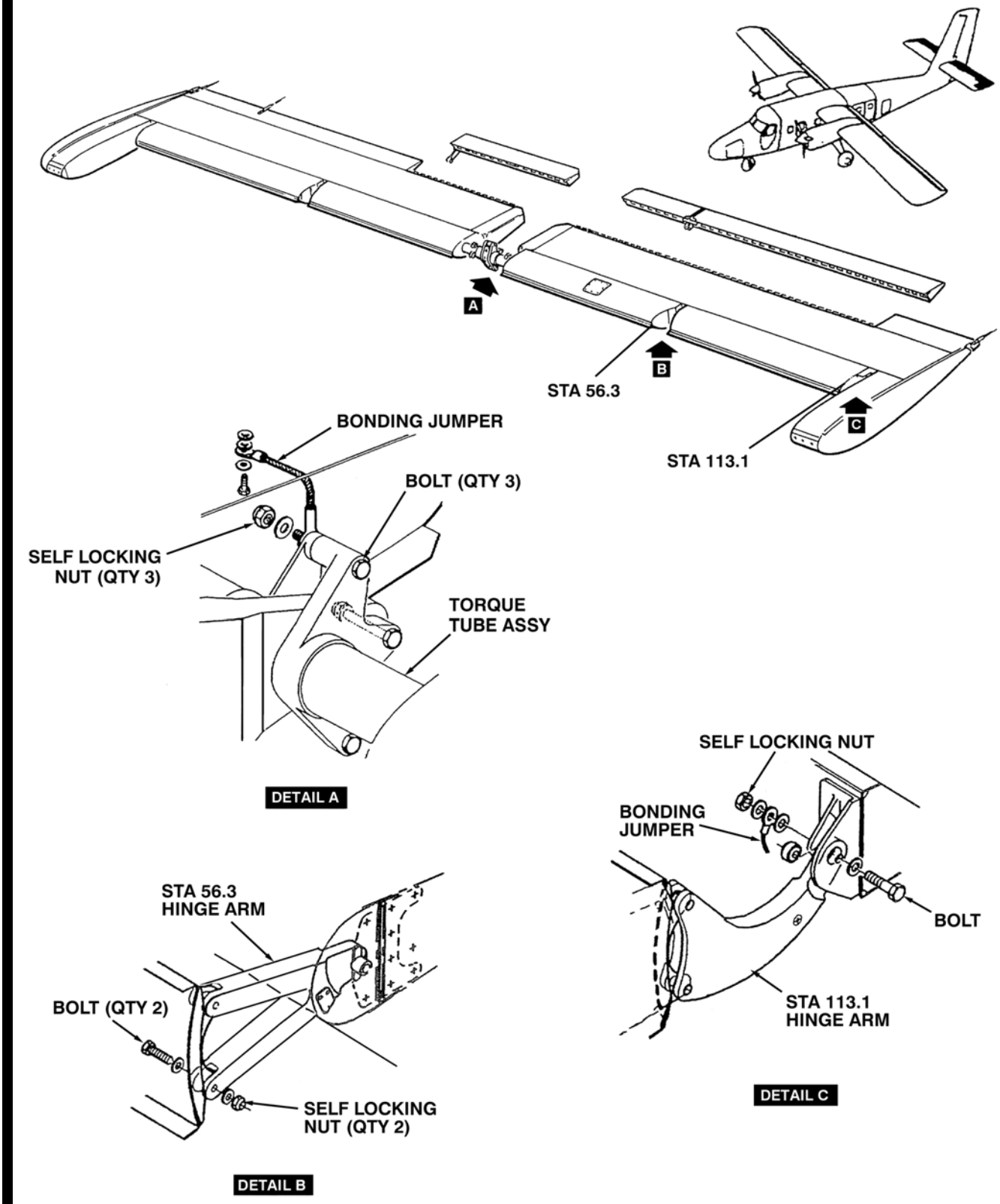
4) Connect flap/elevator interconnect system cables.

5) Connect connecting rod to elevator torque tube.

6) Rig elevator control system (refer to 27-30-00, Elevator Control System – General Data).

7) Do an elevator position operational test as per 27-30-21, Elevator Angle Position Indicator – General Data, Para 3.B.

8) If adjustment is required, adjust the elevator angle position indicator as per 27-30-21, Elevator Angle Position Indicator – General Data, Para 3.A.



Elevator – Removal/Installation
Figure 1

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